rag -- 2026-10-27

5 new paper(s) · topic: Retrieval-augmented generation: retrieval quality, grounding and faithfulness, evaluation benchmarks, and RAG system design.

Worth reading in full

[P4] Hallucination-Aware Multi-Agent RAG via Internal Attention Rewards -- P4 provides a concrete, model-agnostic training recipe that directly integrates a faithfulness signal into the RAG policy and reports sizable reductions in hallucination rates alongside consistent accuracy gains, making it the most informative for anyone studying grounding and evaluation of RAG systems.

Themes

Several papers address the tension between retrieval quality and answer faithfulness. P1 shows that tightening conformal-filter thresholds dramatically raises the proportion of fully-supported claims (up to 97 % on TriviaQA) but cuts answer coverage to as low as 17 % (GPT-6 Luna on NQ) [P1]. P4 tackles the same trade-off by adding a hallucination-risk reward derived from internal attention-head norms, which reduces Answer Hallucination Rate by up to 10.74 points while improving EM/F1 across HotpotQA, RAGTruth and 2WikiMultiHopQA [P4]. Both studies therefore highlight that stricter grounding improves factuality but harms availability, yet they differ in mechanism--post-hoc conformal filtering versus integrated RL-based risk penalisation.

Domain-specific adaptation of RAG is explored in P3, which introduces a failure-aware self-training loop that triages self-generated questions into known, solved, context-failure or retrieval-failure categories and applies tailored updates; this yields consistent gains over SimRAG (e.g., +6.4 points on CS-Bench with a 4B backbone) and demonstrates that diagnosing retrieval versus reader failures can boost performance without changing the retriever [P3]. By contrast, P2 extends RAG to multimodal industrial requirements, retrieving text and image candidates separately and classifying them with a vision-language model, achieving an F1 of 0.310 versus 0.114 for a text-only baseline but still suffering from low image-text recall, illustrating that modality expansion introduces new retrieval bottlenecks [P2].

System-design considerations also appear across the set. P5 builds a fully offline RAG pipeline (FAISS index + local LLMs) for tactical air-combat queries, reporting average user-rating scores around 4.0-4.4/5 but noting that retrieval relevance is the weakest link, echoing the retrieval-quality concerns raised in P1, P3 and P4 [P5]. Together, the papers map a landscape where improving grounding (conformal filtering, hallucination-aware rewards, failure-aware training) must be balanced against retrieval coverage, multimodal challenges, and practical deployment constraints.

5 theme sentence(s) checked against the paper summaries.

At a glance

[P1] Adaptive Retrieval, Selective Answers: Evaluating Conformal Factuality in Agentic RAG · relevance 0.50
The paper evaluates post-generation conformal filtering for factuality in agentic retrieval-augmented generation, showing that stricter conformal targets improve support rates but sharply cut answer availability across GPT-6 Luna and Llama 3.1 8B on three QA benchmarks.

[P2] Multimodal Requirements Traceability Link Recovery in Industrial Settings · relevance 0.45
A multimodal RAG-based approach using a vision-language model improves traceability link recovery on industrial requirements with images, achieving F1 = 0.310 versus 0.114 for a text-only baseline, but still struggles with cross-modal retrieval and has limited external validity.

[P3] TriageRAG: Failure-Aware Self-Training for Adapting Retrieval-Augmented Language Models to Specialized Domains · relevance 0.44

TriageRAG is a failure-aware self-training framework that probes a retrieval-augmented LLM on each self-generated question, categorises the failure type, and routes the example to a tailored training signal, yielding consistent gains over SimRAG on 11 specialized-domain QA datasets.

[P4] Hallucination-Aware Multi-Agent RAG via Internal Attention Rewards · relevance 0.42
A hallucination-aware multi-agent RAG system adds an internal attention-head risk reward to improve answer faithfulness and task performance across several backbones and benchmarks.

[P5] Offline Language-Based Support System for Tactical Decision-Making in Air Combat Operations · relevance 0.42
MATE is an offline, retrieval-augmented generation tool that lets pilots query air-combat manuals in natural language, achieving average answer ratings of 4.00-4.40 (out of 5) with the best performance from Mistral NeMo 12B.

[P1] Adaptive Retrieval, Selective Answers: Evaluating Conformal Factuality in Agentic RAG

Muhammad Aimal Rehman, Chi-Kuang Yeh · arXiv 2610.34872v1 · https://arxiv.org/abs/2610.34872v1 · relevance 0.50

TL;DR: The paper evaluates post-generation conformal filtering for factuality in agentic retrieval-augmented generation, showing that stricter conformal targets improve support rates but sharply cut answer availability across GPT-6 Luna and Llama 3.1 8B on three QA benchmarks.

Problem
Retrieval-augmented generation improves grounding but does not guarantee that every generated claim is supported, prompting the need to assess claim-level conformal factuality in an agentic RAG setting where a language-model policy decides whether to issue follow-up retrievals or stop before generation [1][2].

Method
The study implements an agent that can retrieve up to three documents and make up to three policy decisions, then generates a response, decomposes it into atomic claims, and applies a relevance-scoring function (product-and-max aggregation) followed by conformal filtering based on calibrated thresholds [3][4]. Six configurations are evaluated: two models (GPT-6 Luna and Llama 3.1 8B) on three datasets (HotpotQA, Natural Questions, TriviaQA) using 60 fixed questions per configuration, 30 calibration and 30 test questions, 100 seeded splits, and five conformal target levels (80 %-95 %) plus a no-conformal baseline [1][4][5].

Key results
At the most stringent 95 % conformal target, non-empty response coverage varies from 17.23 % (GPT-6 Luna on NQ) to 60.17 % (Llama 3.1 8B on TriviaQA) [5]; claim retention ranges from 10.72 % to 43.16 % [5]; and the fraction of fully supported non-empty outputs spans 89.14 % to 95.56 % [5]. For Llama 3.1 8B, Table 4 reports 95 %-target coverage of 38.73 % (HotpotQA), 21.00 % (NQ), and 60.17 % (TriviaQA); retained-claim percentages of 27.06 %, 11.77 %, and 39.13 % respectively; fully-supported non-empty rates of 97.33 %, 96.27 %, and 97.72 %; and supported-claim fractions of 93.26 %, 91.47 %, and 94.56 % [7]. Higher conformal targets consistently increase observed full-support rates while reducing answer availability, confirming the reliability-informativeness trade-off [1][6].

Limitations
The evaluation uses controlled subsets of only 60 questions per benchmark, which may limit generalizability [7]; the agentic retrieval is capped at three operations and three policy decisions, leaving longer trajectories unexplored [7]; only two language-model families are tested, so results may not extend to other architectures such as Qwen or DeepSeek [7].

Sources

[1] arXiv 2610.34872v1.pdf, arXiv abstract

[2] arXiv 2610.34872v1.pdf, page 1

[3] arXiv 2610.34872v1.pdf, page 3

[4] arXiv 2610.34872v1.pdf, page 4

[5] arXiv 2610.34872v1.pdf, page 6

[6] arXiv 2610.34872v1.pdf, page 7

[7] arXiv 2610.34872v1.pdf, page 9

[P2] Multimodal Requirements Traceability Link Recovery in Industrial Settings

Haoming Huang, Shinpei Hayashi, Kazuya Kohata, Myeongki Kim et al. · arXiv 2610.34503v1 · https://arxiv.org/abs/2610.34503v1 · relevance 0.45

TL;DR: A multimodal RAG-based approach using a vision-language model improves traceability link recovery on industrial requirements with images, achieving F1 = 0.310 versus 0.114 for a text-only baseline, but still struggles with cross-modal retrieval and has limited external validity.

Problem
Industrial requirements often contain traceable images, which text-only traceability methods cannot process directly, and binary classifiers provide only a single precision-recall trade-off without adjustable balance [1]. The authors aim to reduce both missed links and incorrect predictions for multimodal industrial requirements and to let engineers tune the missed-vs-incorrect link balance per tracing task [2].

Method
The study extends a retrieval-augmented generation (RAG) pipeline by retrieving text and image candidates separately, classifying each pair with a vision-language model (VLM), and filtering predictions using verbalized confidence scores [1][2][3]. Candidate selection uses top-k = 4 per modality (four text and four image candidates) following the original Hey et al. design, and the VLM receives the original content (text block or direct visual input) and outputs a yes/no decision plus a confidence tag in the range [0, 1] [3]. Different confidence thresholds generate alternative link subsets. Evaluation is performed on three non-public industrial requirement sets and six public text-only req2req datasets, comparing against a replicated text-only baseline that follows Hey et al.'s approach [2][4].

Key results
On the industrial test sets the multimodal approach attains an F1 score of 0.310, compared with 0.114 for the replicated text-only baseline [1]. It recovers 49 of 171 industrial test links (precision = 0.338, recall = 0.287, F1 = 0.310), which is 32 more correctly recovered links (2.88 ×) than the baseline [7]. Among the 356 industrial gold links, 150 (42.1 %) involve at least one image, and no text-text gold link exists in Set C, highlighting the need for image handling [4]. With a confidence threshold ? = 1.00, the system recovers 21 image-image links and 16 of the 74 cross-modal gold links, demonstrating limited image-text recall [6][7].

Limitations
Link recovery is constrained by failures to retrieve related text requirements for image queries, leading to low image-text recall [1][6]. The verbalized confidence scores are self-reported, cluster near 1.0, and are not calibrated probabilities, which may affect filtering effectiveness [7]. Industrial datasets are non-public, limiting reproducibility, and the study's external validity is restricted to three sets from a single organization and specific modality distributions [7][8]. The end-to-end comparison changes multiple components (input modalities, embedding model, classifier, prompt, parser, confidence output), preventing isolation of individual effects [7]. Only one practitioner labeled each industrial set, so inter-annotator agreement could not be measured [8].

Sources

[1] arXiv 2610.34503v1.pdf, arXiv abstract

[2] arXiv 2610.34503v1.pdf, page 2

[3] arXiv 2610.34503v1.pdf, page 3

[4] arXiv 2610.34503v1.pdf, page 5

[6] arXiv 2610.34503v1.pdf, page 9

[7] arXiv 2610.34503v1.pdf, page 10

[8] arXiv 2610.34503v1.pdf, page 11

[P3] TriageRAG: Failure-Aware Self-Training for Adapting Retrieval-Augmented Language Models to Specialized Domains

Samarth Varshney, Ritu Yadav · arXiv 2610.33776v1 · https://arxiv.org/abs/2610.33776v1 · relevance 0.44

TL;DR: TriageRAG is a failure-aware self-training framework that probes a retrieval-augmented LLM on each self-generated question, categorises the failure type, and routes the example to a tailored training signal, yielding consistent gains over SimRAG on 11 specialized-domain QA datasets.

Problem
Adapting a general-purpose retrieval-augmented generation (RAG) system to a specialized domain such as medicine is hard when only an unlabeled corpus is available, and existing self-training methods (e.g., SimRAG) treat every retained example identically without distinguishing whether the student already knows the answer or whether failures stem from the reader or the retriever [1][2].

Method
TriageRAG builds on SimRAG's two-stage pipeline but modifies Stage-II by probing the student both closed-book and open-book on each self-generated question, triaging the instance into one of four categories (known, solved, context-failure, retrieval-failure), and applying a corresponding training signal, including a parametric route that teaches the reader to answer when retrieval misses [2]. The retriever is kept frozen in the main configuration, and the triage histogram serves as a label-free diagnostic of domain gaps [2][5].

Key results
- Compared with a budget-matched SimRAG, TriageRAG gains 1.2 points on average in medicine, 1.9 points in science, and 6.4 points on CS-Bench with the 4B backbone, and 4.4 points on CS-Bench with the 8B backbone [3].
- It is the best fine-tuned method on average in every domain and ranks best or second-best on 8 of the 11 datasets [3].
- In the medical domain (Table 1), TriageRAG achieves 81.60 % accuracy on PubMedQA (vs. 82.60 % for SimRAG) and 91.26 % on BioASQ, with the three-round version reaching 80.80 % on PubMedQA and 91.26 % on BioASQ [4].
- In the scientific domain (Table 2), TriageRAG (4B) reaches 89.51 % on ARC-C and 95.60 % on SciQ, with the three-round run attaining 89.16 % and 96.00 % respectively; the 8B backbone on CS-Bench attains 68.64 % (vs. 63.70 % for SimRAG) [4].
- TriageRAG is the only method that recovers or exceeds the closed-book level when retrieval hurts the base model (e.g., 87.6 % vs. 87.4 % in medicine, 68.2 % vs. 68.0 % in science, 70.1 % vs. 69.8 % for Qwen3-8B) [3].
- Analysis shows that ~41 % of self-generated medical questions are "KNOWN" (closed-book correct) and that retrieval-failure rates are higher in medicine (~29 %) than in science or CS (~7 %), explaining why SimRAG wastes budget on low-information examples [5].

Limitations
- The same self-generated pool is re-probed across rounds, causing early consumption of informative examples and a plateau in accuracy; regenerating the pool between rounds is left for future work [6].
- Absolute performance numbers are not directly comparable to SimRAG's larger 8B/27B models because this study uses 4B/8B backbones, a 16 k-example Stage-I blend, and a PubMed slice [6].

- The MedXpertQA dataset remains at chance level for all configurations, limiting conclusions for that task [6].
- Reported results include three-seed averages for headline cells but many cells are single runs, and differences below one point on CS-Bench should be treated as ties [6].
- The parametric route teaches the reader to answer rather than to abstain when context is unhelpful, and the negative result for retriever tuning is based on a single recipe [6].

Sources

[1] arXiv 2610.33776v1.pdf, arXiv abstract

[2] arXiv 2610.33776v1.pdf, page 3

[3] arXiv 2610.33776v1.pdf, page 5

[4] arXiv 2610.33776v1.pdf, page 6

[5] arXiv 2610.33776v1.pdf, page 8

[6] arXiv 2610.33776v1.pdf, page 9

[P4] Hallucination-Aware Multi-Agent RAG via Internal Attention Rewards

Runsong Jia, Qian Chang, Mengjia Wu, Ciprian Doru Giurcaneanu et al. · arXiv 2610.35056v1 · https://arxiv.org/abs/2610.35056v1 · relevance 0.42

TL;DR: A hallucination-aware multi-agent RAG system adds an internal attention-head risk reward to improve answer faithfulness and task performance across several backbones and benchmarks.

Problem
Retrieval-augmented generation often produces answers that contain unsupported claims, i.e., hallucinations, because standard task rewards (e.g., EM or F1) do not explicitly enforce faithfulness to the retrieved evidence [1][2].

Method
The framework decomposes RAG into three turn-based agents--query planner (Q), evidence selector (E), and answer generator (G)--each instantiated from a frozen backbone language model with role-specific LoRA adapters and prompts [2]. A lightweight critic, trained once on TruthfulQA truthful/false pairs, computes a hallucination-risk score H from the generator's internal attention-head norm features; this score is combined with the task reward as R = ?Rtask - ?H during cooperative PPO training, while the backbone and critic remain frozen [2][4].

Key results
- On the Qwen3-8B backbone, the method outperforms the strongest multi-agent baseline (MMOA-RAG) by +2.35 points in Recall/Accuracy, +0.95 EM, and +1.29 F1 on average [3].
- Gains are consistent across datasets: +2.07 Recall on RAGTruth, +2.22 Accuracy and +2.34 EM on HotpotQA, and +2.76 Accuracy and +1.89 EM on 2WikiMultiHopQA [3].
- Answer Hallucination Rate (AHR) drops by 10.74 points on Qwen3-8B, 5.62 points on Llama3-8B, and 3.34 points on Llama2-13B relative to baselines [3].
- The TruthfulQA-calibrated critic achieves 75.92 % AUROC on held-out TruthfulQA pairs, indicating a non-random truthfulness signal in the attention-head features [4].

Limitations
The current critic relies only on attention-head norm features, which keeps computation cheap but may miss richer diagnostics such as hidden-state dynamics or structured attention patterns [6]. Moreover, the critic is kept frozen during policy optimization, preventing it from adapting to policy changes; future work could explore adaptive critic-update strategies [6].

Sources

[1] arXiv 2610.35056v1.pdf, arXiv abstract

[2] arXiv 2610.35056v1.pdf, page 3

[3] arXiv 2610.35056v1.pdf, page 6

[4] arXiv 2610.35056v1.pdf, page 7

[6] arXiv 2610.35056v1.pdf, page 9

[P5] Offline Language-Based Support System for Tactical Decision-Making in Air Combat Operations

Joao P. A. Dantas · arXiv 2610.35635v1 · https://arxiv.org/abs/2610.35635v1 · relevance 0.42

TL;DR: MATE is an offline, retrieval-augmented generation tool that lets pilots query air-combat manuals in natural language, achieving average answer ratings of 4.00-4.40 (out of 5) with the best performance from Mistral NeMo 12B.

Problem
Pilots need rapid, reliable access to tactical information that is normally buried in lengthy manuals, but existing NLP assistants rely on cloud services that are unsuitable for air-gapped or classified environments [1][2].

Method
The authors built MATE, an entirely offline system that preprocesses doctrinal texts, creates embeddings, stores them in a FAISS index, and uses local open-weight LLMs (Llama 3.1 8B, Qwen 2.5 14B, Mistral NeMo 12B) for retrieval-augmented generation [1][4]. A Tkinter GUI and FastAPI backend handle user queries, and an optional Whisper-based voice module was disabled during evaluation [4]. Expert validation involved three Brazilian Air Force fighter pilots (average 6.7 years experience, 933 flight hours) answering five tactical questions derived from Chapter 5 of Shaw's "Fighter Combat" manual, with each question answered by all three models [5][6].

Key results
- Answer ratings (mean ± SD, 5-point Likert) per question: Q1 4.67 ± 0.58 (Llama), 4.00 ± 1.00 (Qwen), 4.67 ± 0.58 (Mistral); Q2 4.67 ± 0.58, 4.33 ± 0.58, 4.67 ± 0.58; Q3 3.00 ± 0.00 for all models; Q4 3.33 ± 0.58, 4.00 ± 1.00, 4.67 ± 0.58; Q5 5.00 ± 0.00, 4.67 ± 0.58, 5.00 ± 0.00 [5].
- Average scores across questions: Llama 4.13, Qwen 4.00, Mistral 4.40 [5].
- General tool evaluation (mean ± SD): relevance of retrieval 3.67 ± 0.58, conceptual/terminological accuracy 4.33 ± 1.15, integration of retrieved context 3.67 ± 0.58, clarity 4.33 ± 0.58, response time 5.00 ± 0.00 [6].
- Pilots rated the response time as fully adequate compared with manual search [1][6].

Limitations
The study involved only three pilots, a single publicly available chapter, and fifteen model answers, which the authors note precludes statistical comparisons or reliable inter-rater analysis [6]. Retrieval relevance received the lowest scores, and the 1,200-character segment limit (with a 4,000-character model context) may truncate important passages, limiting answer quality [6]. These factors make the results exploratory rather than conclusive.

Sources

[1] arXiv 2610.35635v1.pdf, arXiv abstract

[2] arXiv 2610.35635v1.pdf, page 1

[4] arXiv 2610.35635v1.pdf, page 3

[5] arXiv 2610.35635v1.pdf, page 4

[6] arXiv 2610.35635v1.pdf, page 5